

Technology Stack

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Lightweight, fast to build, no heavy framework overhead needed for a simple input-and-result form
2	Backend / Server-Side	Python (Flask)	Simple to integrate directly with scikit-learn models; team already has Python/ML proficiency
3	Machine Learning	scikit-learn (KNN, Logistic Regression, K-Means, Decision Tree, Random Forest)	Well-documented, production-ready implementations of the required classification/clustering algorithms
4	Data Handling	pandas, numpy	Efficient dataset loading, cleaning, and numerical operations for preprocessing and EDA
5	Visualization	matplotlib, seaborn	Used during EDA for univariate, bivariate, and multivariate analysis of soil/climate features
6	Model Persistence	pickle / joblib	Saves the best-performing trained model for fast reuse without retraining on every request
7	Version Control & CI/CD	Git, GitHub	Enables team collaboration, version history, and structured repository organization
8	Development Environment	Jupyter Notebook, VS Code / PyCharm	Notebook for EDA/model experimentation; IDE for backend/app development